Explore the Best Bayesian Model for Dynamic Emotional Network

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GitHub Repo

1 Literature Review

Emotions play a central role in psychological well-being and daily functioning, and understanding how they change over time has become an important topic in psychology and behavioral science Houben et al., 2015; Kuppens, 2017; Pirla et al., 2023. Traditional affect research has often relied on static summaries, such as average questionnaire-based measures of positive or negative affect. However, such approaches overlook temporal dynamics. In reality, emotional states fluctuate continuously and are shaped by both internal processes and external factors, motivating recent work that treats emotions as dynamic systems in which past states influence future states Haslbeck et al., 2022; Kuppens, 2017.

Recent work in the experience sampling methodology (ESM) literature has emphasized the importance of modeling emotional dynamics in daily life. For example, Kullar et al., 2024 examines how emotion measurements evolve over time and proposes improved measurement designs to reduce reporting noise and enhance reliability. However, such approaches typically treat observed emotion scores as direct measures of the underlying emotional state. This assumption may be restrictive, as psychological measurements are often subject to substantial error: self-reported ratings may reflect both the true latent state and additional reporting noise.

To address this limitation, we adopt a Bayesian latent-variable framework in which the true emotional state is modeled as an unobserved continuous variable and the observed ratings are treated as noisy realizations conditional on that state. This accounts for measurement error and provides a more principled generative interpretation of the data. Using the ESM dataset, we compare three increasingly rich models: a naive baseline with independent latent states, a dynamic latent-state model with autoregressive structure, and an extended model with additional lagged emotional predictors.

We also compare two posterior approximation methods: Hamiltonian Monte Carlo (HMC), which yields asymptotically exact posterior samples, and variational inference (VI), which provides a faster but approximate alternative. This comparison highlights the trade-off between computational efficiency and approximation accuracy in modeling emotional dynamics. Accordingly, we aim to identify the most suitable Bayesian model for dynamic emotional networks while comparing HMC and VI in terms of computational efficiency and approximation accuracy.

2 Exploratory Data Analysis

2.1 Choosing the Initial Focal Emotion

Before fitting the Bayesian models, we conducted a brief screening analysis to choose a focal emotion for the univariate naive model. First, we examined the Spearman correlation heatmap for the nine emotion variables (Figure 2 in the appendix). The heatmap showed that `happy` was strongly associated with several positive emotions, especially `pleased` and `enthusiastic`, and moderately negatively associated with several negative emotions, suggesting that it occupies a relatively central position in the emotion system.

We then performed a one-step-ahead screening analysis using leave-one-subject-out cross-validation, since correlation alone does not fully capture predictive usefulness. As shown in Table 2 in the appendix, `happy` ranked first, with the largest mean relative reduction in prediction error. Taken together, these results motivated our choice of `happy` as the focal emotion for the univariate naive Bayesian model.

2.2 Temporal Dependence of Happy

To assess temporal dependence in `happy`, we examined the autocorrelation function (ACF) of its average series over the ordered day-by-beep index. Figure 1 shows positive spikes at lags 5, 10, and 15, suggesting a recurring day-level pattern rather than an independent sequence.

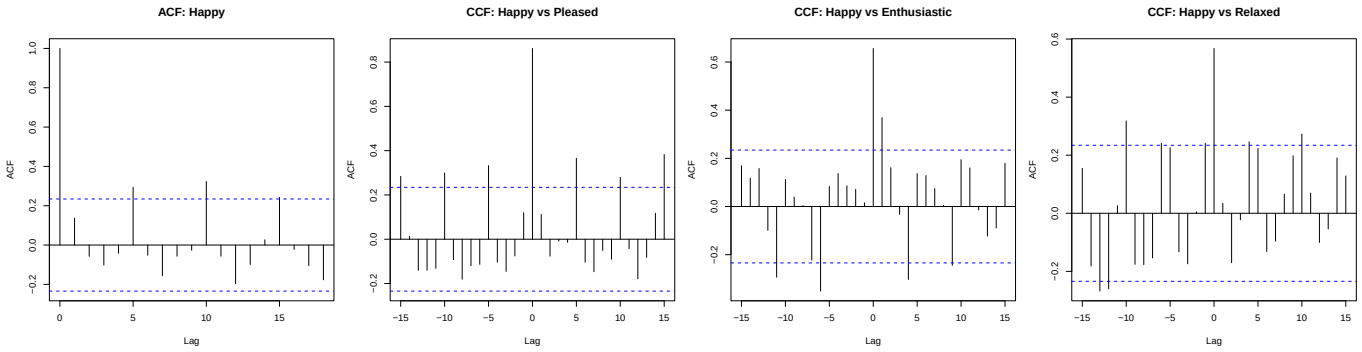


Figure 1: Autocorrelation of `happy` and cross-correlations of `happy` with `pleased` ,`enthusiastic` and `relaxed`.

We also examined the cross-correlation function (CCF) between `happy` and the related positive emotions: `pleased`, `relaxed`, and `enthusiastic`. All pairs show there is a strongest positive association near lag 1, suggesting all three of them are reasonable indicators for `happy`.

3 Methodology

3.1 Models

3.1.1 Naive Model (See Stan code 1)

As a baseline, we use an independent latent-variable model with an ordered logistic likelihood. For each observation $i = 1, \ldots, N$, let $y_i \in \{1, 2, 3, 4, 5, 6, 7\}$ denote the observed ordinal happiness rating

and let $x_i \in \mathbb{R}$ denote a latent continuous happiness level. We assume

$$x_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2), \qquad \mu \sim \mathcal{N}(0, 1.5^2), \qquad \sigma \sim \text{Exponential}(1).$$

Conditional on x_i, the response follows an ordered logistic model,

$$y_i \mid x_i \sim \text{OrderedLogistic}(x_i, \mathbf{c}),$$

where the cutpoints are fixed at $\mathbf{c} = (-2.5, -1.5, -0.5, 0.5, 1.5, 2.5)$. These cutpoints place 0 at the center of the latent scale, corresponding to the middle of the 7-point response scale. Equivalently, the model can be written using a latent logistic noise term:

$$y_i = \begin{cases} 1, & x_i + \epsilon_i \le c_1, \\ 2, & c_1 < x_i + \epsilon_i \le c_2, \\ \vdots & \\ 7, & x_i + \epsilon_i > c_6, \end{cases} \qquad \epsilon_i \sim \text{Logistic}(0, 1).$$

This naive model ignores temporal dependence, subject-specific dynamics, and cross-emotion effects, and instead treats all observations as exchangeable draws from a common latent distribution.

3.1.2 Intermediate Model (with Time Structure) (See Stan code 2)

We next extend the naive model by allowing the latent happiness process to evolve over time. Let x_i again denote the latent continuous happiness state at observation i. For observations with available lagged values, we assume

$$x_i \mid x_{i-1}, x_{i-5} \sim \mathcal{N}(\mu_i, \sigma^2), \qquad \mu_i = \alpha + \beta_1 x_{i-1} + \beta_5 x_{i-5}.$$

Thus, the current latent happiness level depends on both the immediately previous latent state and the latent state five prompts earlier. For early observations where the required lags are unavailable, we instead assume

$$x_i \sim \mathcal{N}(\alpha, \sigma^2), \qquad \alpha \sim \mathcal{N}(0, 1.5^2), \qquad \beta_1, \beta_5 \sim \mathcal{N}(0, 1), \qquad \sigma \sim \text{Exponential}(1).$$

The observation model remains the same as in the naive model. Unlike the naive model, this specification preserves the ordinal likelihood while introducing temporal dependence in the latent process. Missing observed happiness ratings are not removed from the time series; instead, the latent process is defined over all time points and the likelihood is applied only to observed responses, thereby preserving the lag structure.

3.1.3 Extended Model (with Time Structure and Additional Predictors) (See Stan code 3)

Finally, we consider an extension of the intermediate model that incorporates additional lagged emotion predictors. Based on the exploratory screening analysis, the selected predictors were `enthusiastic`, `pleased`, and `relaxed`. Let e_{i-1}, p_{i-1}, and r_{i-1} denote their lagged values at the previous prompt. We then model the latent happiness state as

$$x_i \mid x_{i-1}, x_{i-5}, e_{i-1}, p_{i-1}, r_{i-1} \sim \mathcal{N}(\mu_i, \sigma^2), \qquad \mu_i = \alpha + \beta_1 x_{i-1} + \beta_5 x_{i-5} + \gamma_1 e_{i-1} + \gamma_2 p_{i-1} + \gamma_3 r_{i-1}.$$

Here, β_1 and β_5 capture autoregressive dependence in latent happiness, while $\gamma_1, \gamma_2, \gamma_3$ measure the contributions of lagged `enthusiastic`, `pleased`, and `relaxed`, respectively. For observations where lagged values are unavailable, we again assume $x_i \sim \mathcal{N}(\alpha, \sigma^2)$, with priors

$$\alpha \sim \mathcal{N}(0, 1.5^2), \quad \beta_1, \beta_5 \sim \mathcal{N}(0, 1), \quad \gamma_1, \gamma_2, \gamma_3 \sim \mathcal{N}(0, 1), \quad \sigma \sim \text{Exponential}(1).$$

The observation model remains the same as in the previous models. This model allows future happiness ratings to be explained jointly by persistence in the latent happiness process and by lagged values of related positive-emotion variables.

3.2 Posterior Computation Methods

For each of the three models above, posterior inference is carried out in `Stan` using Hamiltonian Monte Carlo (HMC) and Variational Inference (VI).

Our primary method is HMC, specifically the No-U-Turn Sampler (NUTS), which samples from the exact posterior up to Monte Carlo error. We also consider VI as a faster approximation, using a mean-field variational family that assumes independence among all parameters and latent variables. Although computationally efficient, this approximation may underestimate posterior uncertainty when strong posterior dependence is present.

To compare the two methods directly, we draw 3000 posterior samples from each. For HMC, these are post-warmup Monte Carlo samples; for VI, they are samples from the optimized variational approximation. This enables consistent comparison of posterior summaries and predictive performance.

4 Model and Posterior Comparison Results

To compare both model performance and the accuracy of posterior approximation methods, we evaluate predictive performance using approximate leave-one-out cross-validation (LOO). This approach provides a principled, data-efficient measure of out-of-sample predictive ability without requiring an explicit train–test split.

LOO estimates the expected log predictive density (ELPD), defined as $\text{ELPD} = \sum_{i=1}^{N} \log p(y_i \mid y_{-i})$, where y_{-i} denotes the dataset with the i-th observation removed.

We compare models using differences in ELPD, along with their standard errors, as returned

by `loo_compare`. A large difference relative to its standard error indicates a meaningful difference in predictive performance. This framework enables a unified comparison across models of varying complexity (e.g., with or without temporal structure) and across posterior computation methods (HMC vs. variational inference). See Table 1 for the full comparison.

Table 1: Approximate leave-one-out cross-validation results for all models and posterior computation methods. Higher ELPD indicates better out-of-sample predictive performance, while lower LOOIC indicates better predictive fit. The quantity p_{loo} is the effective number of parameters.

Model	Method	ELPD_{loo} (SE)	p_{loo} (SE)	LOOIC (SE)
Naive	HMC	−12490.6 (24.4)	1179.7 (12.3)	24981.2 (48.8)
Naive	VI	−12570.8 (29.4)	789.6 (8.5)	25141.7 (58.8)
Intermediate	HMC	−10597.5 (32.5)	645.5 (8.3)	21194.9 (65.0)
Intermediate	VI	−11068.1 (31.4)	786.0 (9.1)	22136.1 (62.8)
Extended	HMC	−10559.3 (37.5)	284.9 (3.8)	21118.5 (75.0)
Extended	VI	−10708.0 (42.6)	19.2 (0.2)	21416.0 (85.2)

Across model classes, predictive performance improves as model complexity increases: the naive model performs worst, the intermediate model improves substantially after incorporating latent time structure, and the extended model achieves the highest ELPD after adding lagged emotional predictors. This suggests that both temporal dynamics and cross-emotion covariates help explain variation in observed happiness. Within each model, HMC consistently outperforms VI, with the gap becoming much larger for the intermediate and extended models. This indicates that mean-field VI struggles to capture the stronger posterior dependence induced by temporal structure and latent dynamics. The ELPD differences are also large relative to their standard errors, suggesting that these gaps are meaningful rather than due to Monte Carlo variability.

5 Model Diagnostics for HMC

5.1 Naive Model

We assessed the HMC fit for the naive model using trace plots, rank histograms, an exact invariance test, and standard Monte Carlo efficiency measures. As shown in Figure 3, the trace plots for μ and σ fluctuate within stable ranges with close overlap between chains and no visible drift. The corresponding rank histograms are approximately uniform, suggesting satisfactory mixing.

We also performed an exact invariance test as a debugging check. For both μ and σ, the test did not detect a significant difference between the forward-only and forward-plus-MCMC distributions, suggesting no obvious implementation problem in the hand-written MCMC code.

Finally, we examined Monte Carlo efficiency using $\hat{R}$, bulk ESS, tail ESS, and MCSE. For both μ and σ, $\hat{R} = 1.00$, the effective sample sizes are large, and the Monte Carlo standard errors are very small relative to the posterior standard deviations. Taken together, these results support a stable HMC fit for the naive model.

5.2 Intermediate Model

We assessed the HMC fit for the intermediate model using the same diagnostic tools as for the naive model. As shown in Figure 4, the trace plots for α, β_1, β_5, and σ remain stable with close overlap between chains and no visible drift. The corresponding rank histograms are also approximately uniform, indicating satisfactory mixing.

As shown in Table 6, the exact invariance test did not indicate any meaningful discrepancy between the forward-only and forward-plus-MCMC distributions. This provides further support that the hand-written MCMC implementation does not show an obvious debugging issue.

We also evaluated Monte Carlo efficiency in Table 7. All four key parameters had $\hat{R} = 1.00$, large effective sample sizes, and negligible Monte Carlo error relative to posterior uncertainty. Overall, these diagnostics support a stable HMC fit for the intermediate model.

5.3 Extended Model

We assessed the HMC fit for the extended model using trace plots and rank histograms for the main parameters α, β_1, β_5, γ_{enth}, γ_{pleased}, γ_{relaxed}, and σ. As shown in Figures 5 and 6, the trace plots remain within stable ranges with close overlap between chains and no obvious drift. The rank histograms are broadly consistent with approximate uniformity, suggesting acceptable mixing.

As shown in Tables 8 and 9, the extended model did not show any obvious diagnostic problems. The exact invariance test did not detect a meaningful difference between the forward-only and forward-plus-MCMC distributions for any of the selected summaries. In addition, all key parameters had $\hat{R} = 1.00$, with reasonably large effective sample sizes and small Monte Carlo standard errors. Overall, these results support a stable HMC fit for the extended model.

6 Conclusion and Discussion

Overall, the results show that incorporating temporal structure and additional predictors substantially improves predictive performance, with the extended model—combining time dependence and lagged emotional variables—achieving the best predictive accuracy and therefore providing the most effective Bayesian model for capturing dynamic emotional processes. At the same time, Hamiltonian Monte Carlo (HMC) consistently delivers a more reliable posterior approximation than variational inference with satisfactory diagnostic checks results.

Despite these strengths, several limitations should be noted. First, mean-field variational inference is a restrictive approximation, especially for time-dependent latent models where strong posterior dependence is expected. This likely contributes to the poorer predictive performance of VI relative to HMC. Future work could explore richer variational approximations that better capture posterior dependence. Second, although approximate leave-one-out cross-validation provides a useful measure of predictive performance, it does not fully respect the temporal ordering of the data. More appropriate time-series validation schemes, such as rolling-origin evaluation, could be considered in future work.

7 Author Contributions

This project was completed through a collaborative effort between the two authors, with responsibilities divided to ensure both efficiency and consistency across different components of the study.

Yu Chang was primarily responsible for conducting the literature review and drafting the introduction and background sections, including the theoretical motivation for the modeling approaches considered. Marc Shi was primarily responsible for the methodological development, implementation of the models in `Stan`, and the analysis of results, including posterior diagnostics and predictive performance evaluation.

For the computational component, both authors independently implemented and ran different model specifications, which enabled cross-validation of results and ensured the robustness of the computational workflow. The final results, including model comparisons and tables, were compiled and organized by Marc Shi.

Throughout the project, both authors collaborated closely by discussing modeling choices, validating intermediate results, and refining the structure and presentation of the report. Both authors contributed to editing and reviewing the final manuscript to ensure clarity, accuracy, and coherence.

8 appendix

8.1 Spearman Correlation Heatmap

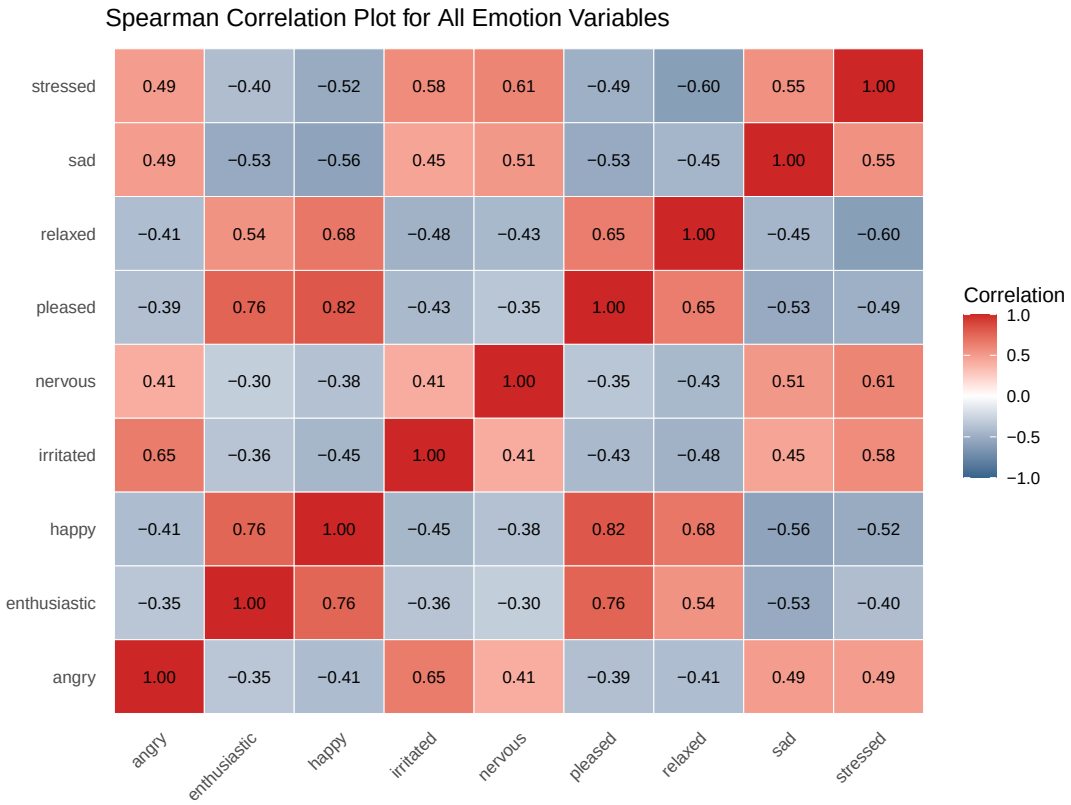


Figure 2: Spearman correlation heatmap for the nine emotion variables.

8.2 Leave-One-Subject-Out Predictive Screening for Variable Selection

To rank the candidate emotions by predictive usefulness, we carried out a leave-one-subject-out one-step-ahead prediction analysis. For each ordered predictor–target pair (j, k) with $j \neq k$, we compared a baseline model and an augmented model:

$$\text{Baseline model:} \quad Y_{i,t+1}^{(k)} = \alpha_k + \beta_k Y_{i,t}^{(k)} + \varepsilon_{i,t}^{(k)}, \tag{1}$$

$$\text{Augmented model:} \quad Y_{i,t+1}^{(k)} = \alpha_k + \beta_k Y_{i,t}^{(k)} + \gamma_{jk} Y_{i,t}^{(j)} + \varepsilon_{i,t}^{(k)}. \tag{2}$$

Here, the baseline model predicts the next value of target emotion k using only its own lagged value, while the augmented model additionally includes predictor emotion j at the previous time point. Thus, the augmented model tests whether emotion j provides incremental predictive information beyond the target emotion's own temporal dependence.

Prediction performance was evaluated using leave-one-subject-out cross-validation. In each fold, one subject was held out as the test set, the models were fitted on the remaining subjects, and prediction error was computed on the held-out subject. This was repeated for every subject and then aggregated.

For each predictor emotion j, we summarized its overall predictive usefulness across all other target emotions by

$$\text{Score}(j) = \frac{1}{K-1} \sum_{k \neq j} \frac{\text{MSE}_{\text{baseline},k} - \text{MSE}_{\text{augmented},k}}{\text{MSE}_{\text{baseline},k}}, \tag{3}$$

where $K = 9$ is the number of emotions. This score is the mean relative reduction in out-of-sample mean squared error across all other target emotions. Larger values indicate stronger average incremental predictive value.

Table 2 reports the resulting summary statistics. The column `total_rmse_gain` is the total reduction in RMSE across all target emotions, and `mean_rmse_gain` is the corresponding average reduction. The primary ranking criterion is `mean_rel_mse_reduction`, which corresponds directly to the score above. Under this criterion, `happy` ranked first, indicating that it provided the largest average out-of-sample improvement when added as a lagged predictor for the remaining emotions. The column `median_rel_mse_reduction` gives a more robust summary across targets and also placed `happy` first.

8.3 Forward Selection for Additional Predictors of `happy`

We used a forward-selection procedure to identify additional emotion variables that help predict future `happy`. At each step, we started from a baseline model containing $\texttt{happy}_{i,t}$, then added one candidate emotion at time t, and selected the candidate that gave the largest improvement in one-step-ahead prediction of $\texttt{happy}_{i,t+1}$.

Prediction performance was evaluated using leave-one-subject-out cross-validation. In each fold, one subject was held out as the test set, the model was fitted on the remaining subjects, and prediction

Table 2: Overall predictive screening scores from leave-one-subject-out cross-validation. Emotions were ranked primarily by mean relative MSE reduction.

Predictor	Total RMSE Gain	Mean RMSE Gain	Mean Relative MSE Reduction	Median Relative MSE Reduction
happy	0.026735	0.003342	0.005293	0.004320
enthusiastic	0.017941	0.002243	0.003581	0.003008
relaxed	0.015705	0.001963	0.003187	0.003463
pleased	0.012549	0.001568	0.002535	0.002532
nervous	0.010544	0.001318	0.002081	0.001074
stressed	0.008082	0.001010	0.001651	0.001179
sad	0.007534	0.000942	0.001546	0.001556
irritated	0.001180	0.000147	0.000307	-0.000136
angry	-0.003500	-0.000438	-0.000716	-0.000738

error was computed on the held-out subject. Candidates were ranked by their relative reduction in out-of-sample mean squared error,

$$\frac{\mathrm{MSE}_{\mathrm{base}} - \mathrm{MSE}_{\mathrm{aug}}}{\mathrm{MSE}_{\mathrm{base}}},$$

where larger values indicate stronger incremental predictive value.

Table 3 shows the selected predictors at each step. The first three selected variables were `enthusiastic`, `relaxed`, and `pleased`. The predictive gain was largest at the first step and became smaller afterwards, suggesting diminishing additional benefit as more closely related positive emotions were added.

Table 3: Forward-selection results for predicting future `happy`.

Step	Candidate	RMSE Base	RMSE Aug	RMSE Gain	Relative MSE Reduction	n_{obs}	Selected So Far
1	enthusiastic	1.165833	1.161850	0.003984	0.006822	5941	enthusiastic
2	relaxed	1.161850	1.160139	0.001711	0.002943	5941	enthusiastic, relaxed
3	pleased	1.160221	1.159928	0.000293	0.000505	5940	enthusiastic, relaxed, pleased

8.4 Model Diagnosis Figures and Tables

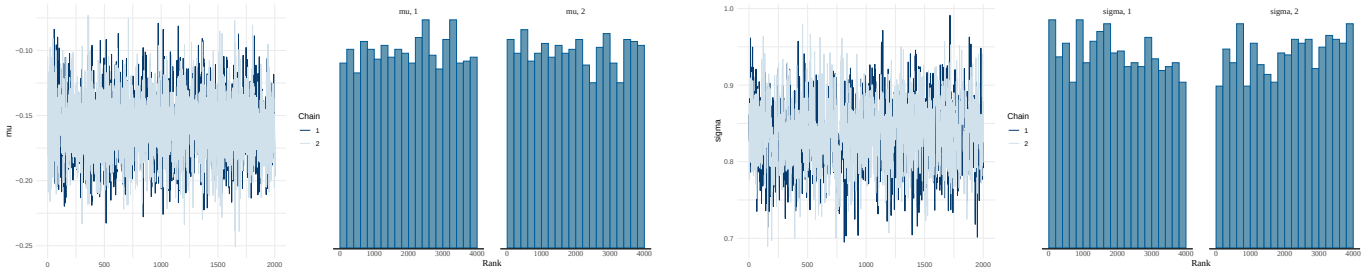
8.4.1 Naive Model

Table 4: Exact invariance test results for the naive model.

Summary	D-statistic	p-value
μ	0.075	0.627
σ	0.090	0.393

Table 5: Monte Carlo efficiency diagnostics for the naive model.

Parameter	Mean	SD	MCSE Mean	ESS Bulk	ESS Tail	$\hat{R}$
μ	-0.158	0.0245	0.000311	6207	2951	1.00
σ	0.837	0.0429	0.00126	1148	1943	1.00



(a) mu diagnostics

(b) sigma diagnostics

Figure 3: HMC diagnostic plots for the naive model.

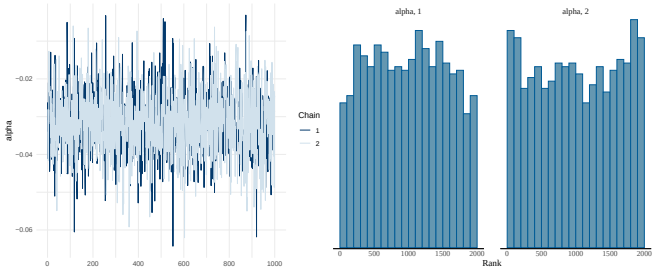
8.4.2 Intermediate Model

Table 6: Exact invariance test results for the intermediate model.

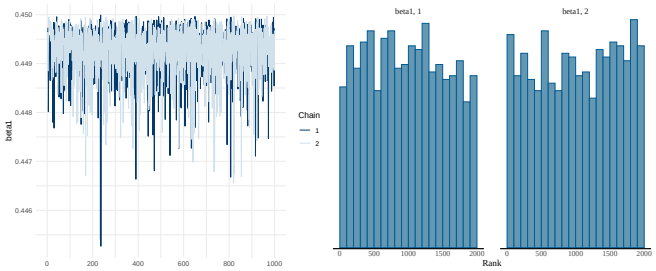
Summary	D-statistic	p-value
α	0.105	0.220
β_1	0.060	0.864
β_5	0.045	0.987
σ	0.075	0.627

Table 7: Monte Carlo efficiency diagnostics for the intermediate model.

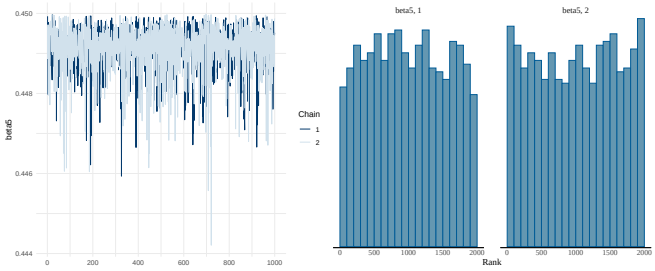
Parameter	Mean	SD	MCSE Mean	ESS Bulk	ESS Tail	$\hat{R}$
α	-0.0316	0.00859	0.000198	1881	1430	1.00
β_1	0.449	0.000560	0.0000101	2815	1235	1.00
β_5	0.449	0.000627	0.0000113	2984	1313	1.00
σ	0.680	0.0170	0.000412	1716	1268	1.00



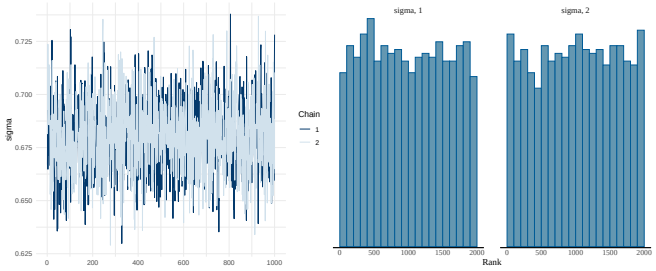
(a) Alpha diagnostics



(b) Beta$_1$ diagnostics



(c) Beta$_5$ diagnostics



(d) Sigma diagnostics

Figure 4: HMC diagnostic plots for the intermediate model.

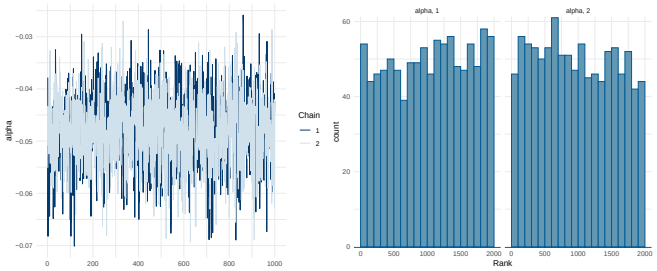
8.4.3 Extended Model

Table 8: Exact invariance test results for the extended model.

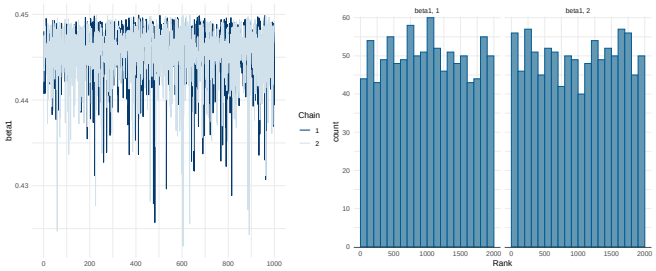
Summary	D-statistic	p-value
α	0.045	0.987
β_1	0.100	0.270
β_5	0.055	0.923
γ_{enth}	0.095	0.327
γ_{pleased}	0.075	0.627
γ_{relaxed}	0.050	0.964
σ	0.060	0.864

Table 9: Monte Carlo efficiency diagnostics for the extended model.

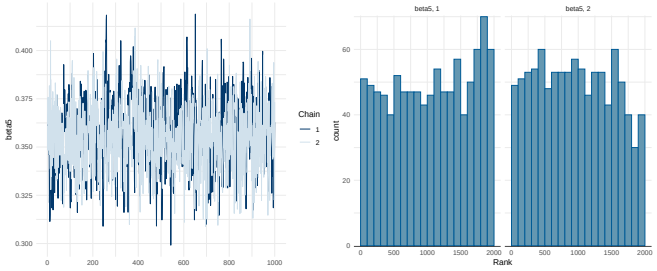
Parameter	Mean	SD	MCSE Mean	ESS Bulk	ESS Tail	$\hat{R}$
α	-0.0480	0.00708	0.000166	1810	1328	1.00
β_1	0.445	0.00357	0.000607	2962	1164	1.00
β_5	0.356	0.0179	0.000631	804	940	1.00
γ_{enth}	0.126	0.0238	0.000596	1597	1436	1.00
γ_{pleased}	0.259	0.0299	0.000918	1048	1515	1.00
γ_{relaxed}	0.0990	0.0184	0.000460	1600	1262	1.00
σ	0.380	0.0236	0.000876	726	1199	1.00



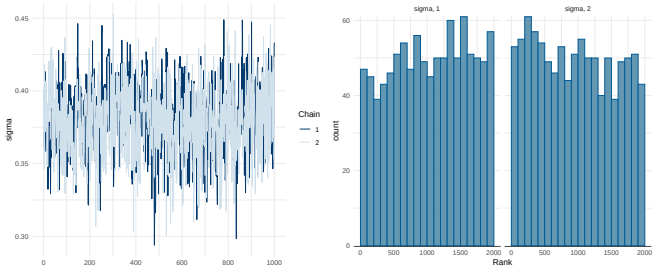
(a) Alpha diagnostics



(b) Beta$_1$ diagnostics

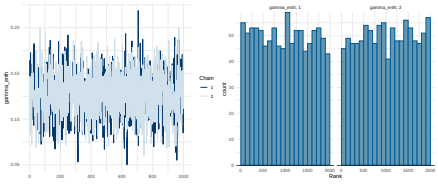


(c) Beta$_5$ diagnostics

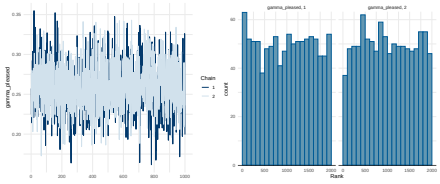


(d) Sigma diagnostics

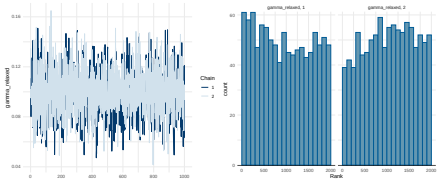
Figure 5: HMC diagnostic plots for α, β_1, β_5, and σ in the extended model.



(a) γ_{enth} diagnostics



(b) γ_{pleased} diagnostics



(c) γ_{relaxed} diagnostics

Figure 6: HMC diagnostic plots for the lagged predictor coefficients in the advanced model.

8.5 Implementation R codes

8.6 Naive Model

```
data {
  int<lower=1> N;
  array[N] int<lower=1, upper=7> y;
}

transformed data {
  vector[6] c;
  c[1] = -2.5;
  c[2] = -1.5;
  c[3] = -0.5;
  c[4] = 0.5;
  c[5] = 1.5;
```

```
  c[6] = 2.5;
}

parameters {
  real mu;
  real<lower=0> sigma;
  vector[N] z;
}

transformed parameters {
  vector[N] x;
  x = mu + sigma * z;
}

model {
  // Priors
  mu ~ normal(0, 1.5);
  sigma ~ exponential(1);
  z ~ normal(0, 1);

  // Likelihood
  y ~ ordered_logistic(x, c);
}

generated quantities {
  array[N] real log_lik;
  array[N] int<lower=1, upper=7> y_rep;

  real x_new;
  int<lower=1, upper=7> y_new;

  for (n in 1:N) {
    log_lik[n] = ordered_logistic_lpmf(y[n] | x[n], c);
    y_rep[n] = ordered_logistic_rng(x[n], c);
  }


  x_new = normal_rng(mu, sigma);
  y_new = ordered_logistic_rng(x_new, c);

}
```

Listing 1: Stan code for the naive model.

8.7 Intermediate Model

```
data {
  int<lower=1> N;
  int<lower=0> N_obs;

  array[N_obs] int<lower=1, upper=7> y_obs;
  array[N_obs] int<lower=1, upper=N> obs_index;

  array[N] int<lower=0, upper=1> has_lag1;
  array[N] int<lower=0, upper=N> lag1_index;

  array[N] int<lower=0, upper=1> has_lag5;
  array[N] int<lower=0, upper=N> lag5_index;
}

transformed data {
  vector[6] c;
  c[1] = -2.5;
  c[2] = -1.5;
  c[3] = -0.5;
  c[4] = 0.5;
  c[5] = 1.5;
  c[6] = 2.5;
}

parameters {
  real alpha_raw;
  real beta1_raw;
  real beta5_raw;
  real<lower=0> sigma;
  vector[N] z;
}

transformed parameters {
  real alpha;
  real beta1;
  real beta5;
  vector[N] mean_x;
  vector[N] x;

```

```
  alpha = alpha_raw;

  // Bound coefficients to a reasonable range
  // 0.45 keeps the recursion from exploding too easily
  beta1 = 0.45 * tanh(beta1_raw);
  beta5 = 0.45 * tanh(beta5_raw);

  for (i in 1:N) {
    mean_x[i] = alpha;

    if (has_lag1[i] == 1)
      mean_x[i] += beta1 * x[lag1_index[i]];

    if (has_lag5[i] == 1)
      mean_x[i] += beta5 * x[lag5_index[i]];

    x[i] = mean_x[i] + sigma * z[i];
  }
}

model {

  alpha_raw ~ normal(0, 1.0);
  beta1_raw ~ normal(0, 1.0);
  beta5_raw ~ normal(0, 1.0);
  sigma ~ exponential(2);
  z ~ normal(0, 1);

  for (n in 1:N_obs) {
    y_obs[n] ~ ordered_logistic(x[obs_index[n]], c);
  }
}

generated quantities {
  array[N_obs] real log_lik;
  array[N_obs] int<lower=1, upper=7> y_rep_obs;

  real x_new;
  int<lower=1, upper=7> y_new;

  for (n in 1:N_obs) {
```

```
    log_lik[n] = ordered_logistic_lpmf(y_obs[n] | x[obs_index[n]], c);
    y_rep_obs[n] = ordered_logistic_rng(x[obs_index[n]], c);
  }

  x_new = normal_rng(alpha, sigma);
  y_new = ordered_logistic_rng(x_new, c);
}
```

Listing 2: Stan code for the intermediate model.

8.8 Extended Model

```
data {
  int<lower=1> N;
  int<lower=0> N_obs;

  array[N_obs] int<lower=1, upper=7> y_obs;
  array[N_obs] int<lower=1, upper=N> obs_index;


  array[N] int<lower=0, upper=1> has_lag1;
  array[N] int<lower=0, upper=N> lag1_index;

  array[N] int<lower=0, upper=1> has_lag5;
  array[N] int<lower=0, upper=N> lag5_index;


  vector[N] enth_lag1;
  vector[N] pleased_lag1;
  vector[N] relaxed_lag1;

  array[N] int<lower=0, upper=1> has_enth_lag1;
  array[N] int<lower=0, upper=1> has_pleased_lag1;
  array[N] int<lower=0, upper=1> has_relaxed_lag1;
}

transformed data {
  vector[6] c;
  c[1] = -2.5;
  c[2] = -1.5;
  c[3] = -0.5;
  c[4] = 0.5;
  c[5] = 1.5;
  c[6] = 2.5;
}

parameters {
  real alpha_raw;
  real beta1_raw;
  real beta5_raw;

```

```
  real gamma_enth;
  real gamma_pleased;
  real gamma_relaxed;

  real<lower=0> sigma;
  vector[N] z;
}

transformed parameters {
  real alpha;
  real beta1;
  real beta5;

  vector[N] mean_x;
  vector[N] x;

  alpha = alpha_raw;


  beta1 = 0.45 * tanh(beta1_raw);
  beta5 = 0.45 * tanh(beta5_raw);

  for (i in 1:N) {
    mean_x[i] = alpha;

    if (has_lag1[i] == 1)
      mean_x[i] += beta1 * x[lag1_index[i]];

    if (has_lag5[i] == 1)
      mean_x[i] += beta5 * x[lag5_index[i]];

    if (has_enth_lag1[i] == 1)
      mean_x[i] += gamma_enth * enth_lag1[i];

    if (has_pleased_lag1[i] == 1)
      mean_x[i] += gamma_pleased * pleased_lag1[i];

    if (has_relaxed_lag1[i] == 1)
      mean_x[i] += gamma_relaxed * relaxed_lag1[i];

    x[i] = mean_x[i] + sigma * z[i];
```

```
  }
}

model {
  alpha_raw ~ normal(0, 1.0);
  beta1_raw ~ normal(0, 1.0);
  beta5_raw ~ normal(0, 1.0);

  gamma_enth ~ normal(0, 0.5);
  gamma_pleased ~ normal(0, 0.5);
  gamma_relaxed ~ normal(0, 0.5);

  sigma ~ exponential(2);
  z ~ normal(0, 1);

  for (n in 1:N_obs) {
    y_obs[n] ~ ordered_logistic(x[obs_index[n]], c);
  }
}

generated quantities {
  array[N_obs] real log_lik;
  array[N_obs] int<lower=1, upper=7> y_rep_obs;

  real x_new;
  int<lower=1, upper=7> y_new;

  for (n in 1:N_obs) {
    log_lik[n] = ordered_logistic_lpmf(y_obs[n] | x[obs_index[n]], c);
    y_rep_obs[n] = ordered_logistic_rng(x[obs_index[n]], c);
  }

  x_new = normal_rng(alpha, sigma);
  y_new = ordered_logistic_rng(x_new, c);
}
```

Listing 3: Stan code for the extended model.

References

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